

Recursion is a programming technique where a method calls itself repeatedly until a base condition is reached.

1. Simplifies repetitive calculations.
2. Makes the code shorter and easier to understand.
3. Useful for mathematical and hierarchical problems.

Time Complexity

The recursive function makes **one recursive call per year**.

If there are **n years**:

Time Complexity = $O(n)$

Space Complexity

Since each recursive call is stored on the call stack:

Space Complexity = $O(n)$

It can be optimised by:

1. Use an **iterative loop** to avoid stack usage.
2. Use the **compound interest formula** directly